

- 6 Fig. 6.1 shows a capacitor of capacitance C connected in series with a resistor of resistance R .

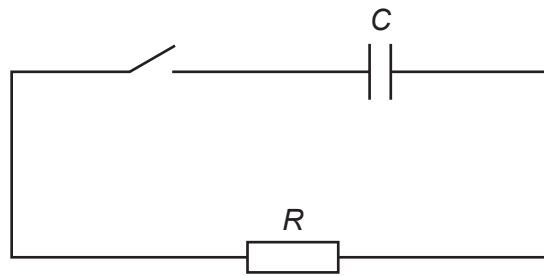


Fig. 6.1

Initially the switch is open and there is a p.d. of 12 V across the capacitor.

At time $t = 0$, the switch is closed so that there is a current I in the resistor.

Fig. 6.2 shows the variation of I with t .

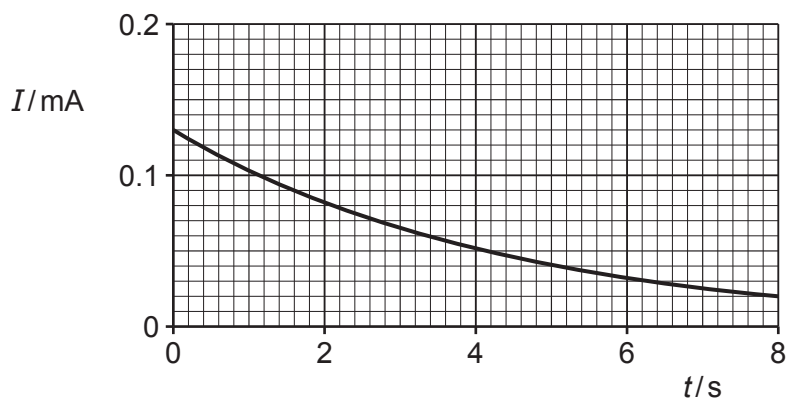


Fig. 6.2

- (a) Explain the shape of the line in Fig. 6.2.

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..... [3]

(b) Use Fig. 6.2 to determine:

(i) resistance R

$$R = \dots\dots\dots \Omega \text{ [2]}$$

(ii) the time constant τ of the circuit in Fig. 6.1.

$$\tau = \dots\dots\dots \text{ s [3]}$$

(c) Use your answers in (b) to determine capacitance C .

$$C = \dots\dots\dots \text{ F [2]}$$

[Total: 10]